TABLETOP / AGENTIC KUBERNETES GUARDRAILS

FROM INTENT TO ENFORCEMENT

Quarantine Mode

Incident containment on a shared Kubernetes platform



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IDENTITY WORKLOAD CONTROL PLANE

Which actions must be restricted?

Which dependencies must remain available?

Record the requirement, enforcement layer and acceptance evidence.

Review a containment change.

Incident lead. Recorder. Analyst.
Independent reviewer.
The independent reviewer states a concern before each decision.

00–06	Briefing, case and constraint choice
06–18	Intent, compilation demonstration, PR review
18–26	Exceptions and rollout
26–38	Containment and late evidence
38–60	Lab consequences, peer review and recovery

Unexpected workload activity

SCOPE UNCERTAIN

Several production services share the cluster.
The affected service remains operational.

Restrict affected identities and workloads while preserving essential dependencies.

Four prepared scenarios

A	**Library update**	A library update and unexpected payments traffic
B	**Support assistant**	A ticket-reading assistant and unexpected tool activity
C	**Unapproved mutation**	An ops agent changes replicas without a ticket
D	**Secret enumeration**	A monitoring identity lists Secrets across namespaces

Operational constraints

Availability	Maintain minimum service capacity and essential dependency access.
Evidence	Preserve evidence and record every approval.
Velocity	Continue safe delivery while affected changes pause.
Containment	Stop suspected exposure and justify the service cost.

Name your minimum acceptable service.

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Core review points

+3

Independent review points

Intent and PR review

Control boundary / availability floor / missing test

Exception and rollout

Verified approver / owner and expiry / narrow scope

Containment

Ordered actions / protected dependency / preserved evidence

Recovery

Exit test / rollback authority / verified return to service

Scores reflect documented reasoning and evidence, not restriction severity.

01

Define the security requirement.

Identify the prohibited behavior and essential dependencies.



Typed decisions and deployment authority

Owner	Responsibility
Decision provider	Choose a typed control, dependency or next action
Code and templates	Fill approved values, validate the spec and render YAML
Tools	Observe workloads and simulate admission outcomes
Human reviewer	Resolve missing controls, approve exceptions and deploy

Intent → reviewable proposal

“Prevent privilege escalation by workloads in the payments namespace.”

Control: deny_privilege_escalation
Templates produce a Kyverno policy. Fixtures show its effect.

The real System-1 model failed the current live-demo gate.
This enforcement demonstration uses a supplied reference spec.

Unresolved enforcement requirements

Intent clause	Proposal status
Limit east-west communication	Egress-profile policy
Prevent privilege escalation	Privilege policy
Restrict Kubernetes API access	Unimplemented. RBAC handover required.
Avoid disruption and expire exceptions	Rollout and exception process

Simulation evidence is limited to the fixtures exercised.

Record your review decision

APPROVE

MODIFY

REJECT

ESCALATE

One control to keep or change.

One missing test that could change your decision.

Exception approval is unverified

The request cites operational need but provides no verified approver or expiry.

VERIFY THE APPROVER. NARROW THE SCOPE. SET THE EXPIRY.

22–26 MINUTES / TEAM TASK

Audit. Warn. Deny.

Mode	Admission behavior	Review question
Audit	Admit and record	Is exposure acceptable while we observe?
Warn	Admit and warn	Who will act on the warning?
Deny	Refuse the change	What legitimate workload could break?

Choose a mode per control and write promotion criteria.

Quarantine Mode

Level	Control change	Operational effect
Q0 Normal	Baseline guardrails	Normal service
Q1 Freeze	Freeze covered agent writes and new exceptions	Running Pods remain
Q2 Constrain	Capture evidence, constrain suspect network and rotation	Clean capacity can serve
Q3 Isolate	Remove remaining suspect network allowances	Degraded service may follow

03

Plan scoped containment.

Specify control changes, evidence and minimum service capacity.



Reassess with additional evidence

REVISE THE PLAN / ESCALATE / JUSTIFY NO CHANGE

Record which new evidence changes the scope or action order.

Validate the containment assumptions

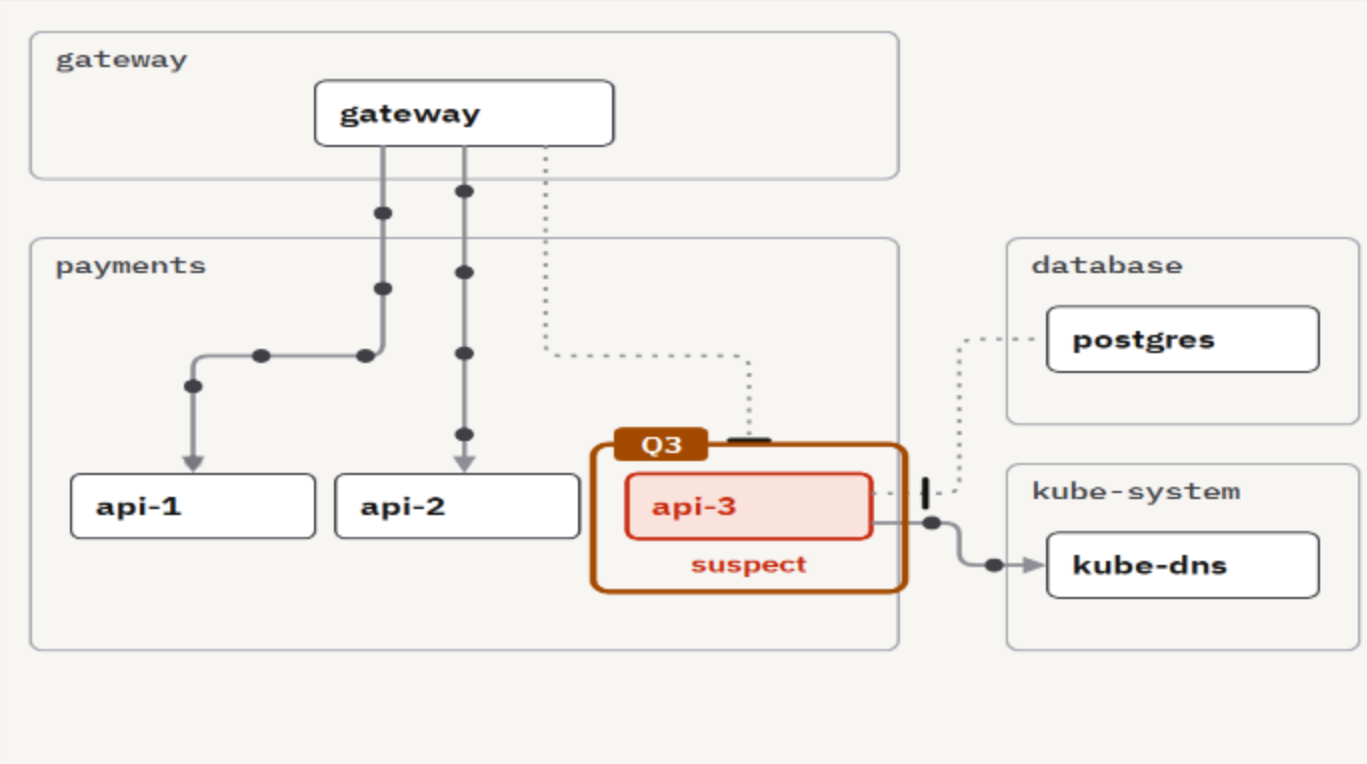
NETWORK Does deny-all remove an existing allow?

RECOVERY Can eviction preserve evidence and healthy capacity?

AUTHORITY Which identity or API path can bypass your plan?

Real k3s observations. Fictional incident. Recorded, not live.

Is the suspect isolated?

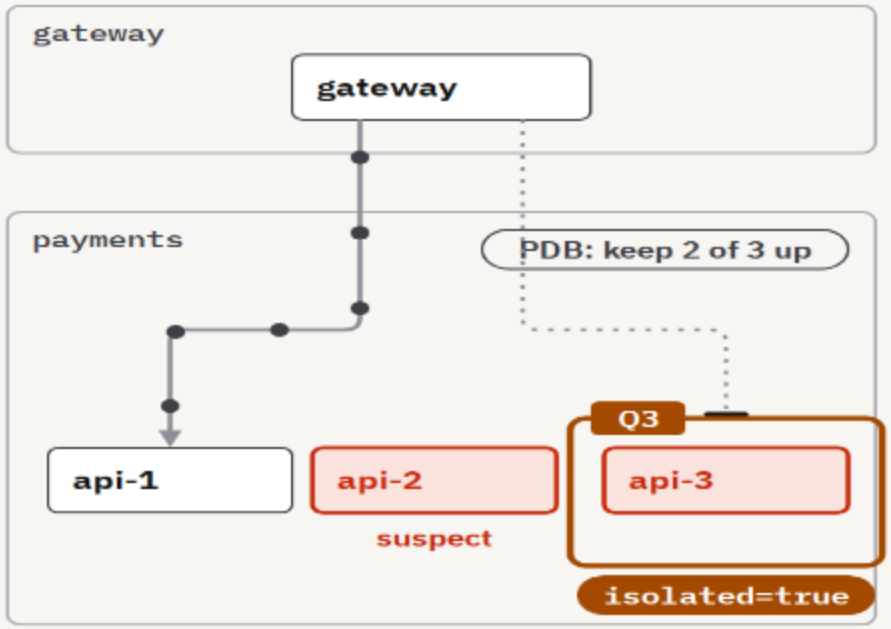


Q3 is declared.
Look at the DNS path.

RECORD YOUR DECISION. STATE THE SUPPORTING EVIDENCE.

Recorded k3s observations; simplified workload names. Scenario context is fictional.

Can both suspect Pods be evicted?

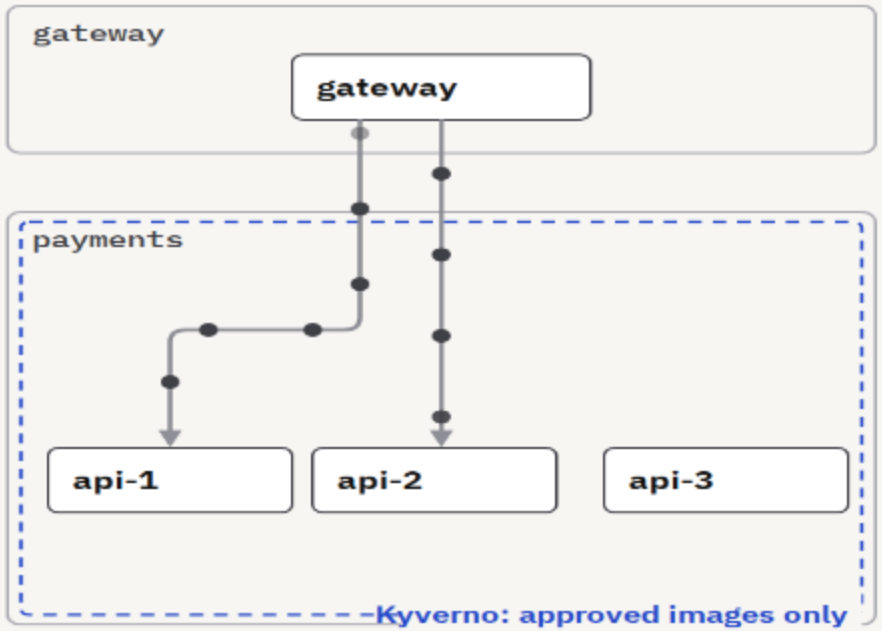


Three replicas.
The PDB keeps two up.

RECORD YOUR DECISION. STATE THE SUPPORTING EVIDENCE.

Recorded k3s observations; simplified workload names. Scenario context is fictional.

Does Q2 cover the debug request?

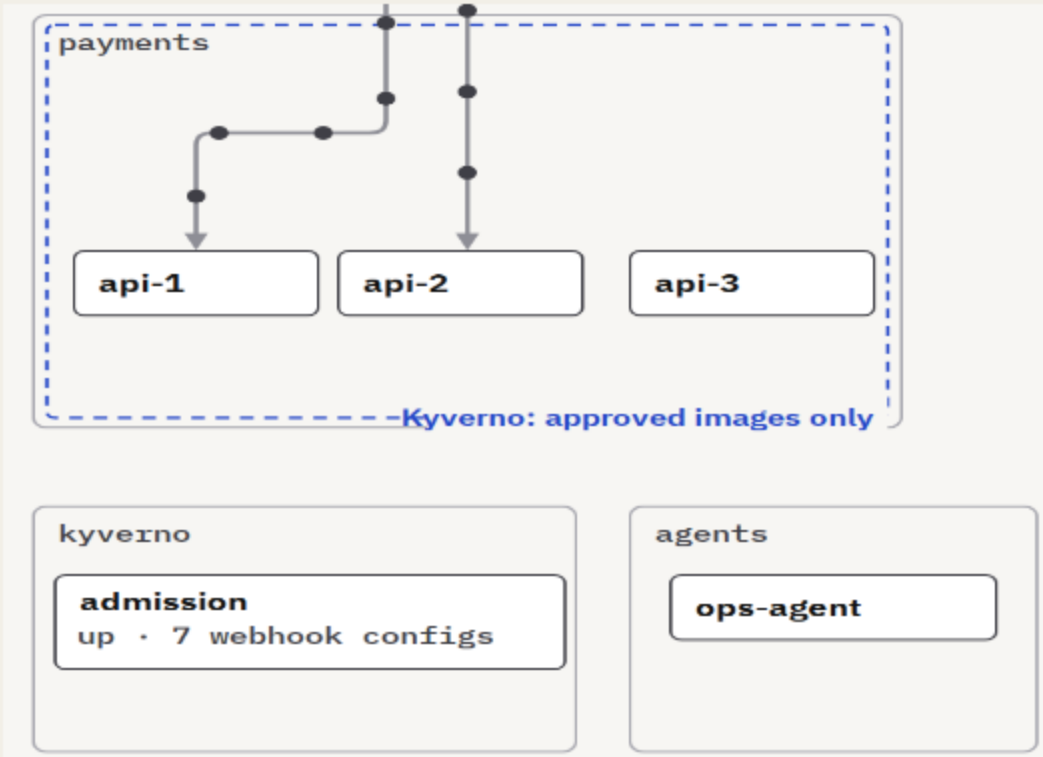


Only approved images may start in payments.

RECORD YOUR DECISION. STATE THE SUPPORTING EVIDENCE.

Recorded k3s observations; simplified workload names. Scenario context is fictional.

Can this identity disable admission?



Its role can scale any Deployment.

RECORD YOUR DECISION. STATE THE SUPPORTING EVIDENCE.

Recorded k3s observations; simplified workload names. Scenario context is fictional.

Review findings and unresolved requirements

Case	What changes the response
A	The updated image is present on multiple replicas
B	The assistant follows injected instructions through tools
C	A human executes changes outside the agent identity scope
D (lab)	Copied credential still reads Secrets after Pod quarantine

Trained model proposal

Requirement	Selected control / routing	Score
Network restriction	Privilege escalation	0.8986
Privilege escalation	Privilege escalation	0.8574
Kubernetes API access	Human review	0.5895
Rollout and expiry	Privilege escalation	0.9107

One privilege policy. Six privilege tests pass.

Which requirements lack enforcement evidence?

Passing privilege tests leave the proposal incomplete

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Pod CREATE privilege tests

Network	Wrong control selected
API access	Human disposition required
Rollout / expiry	Wrong control selected

Record the enforcement owner and the evidence required to proceed.

MEASURED PILOT / THREE-RUN WORKFLOW MEDIANS

Model and workflow evaluation

JEV 1.13	CLAUDE OPUS 5.5	LAYA / ZERO-SHOT
30/33	**32/33**	**16/33**
1.82 s / run	**24.18 s / run**	**0.41 s / run**
$0.00019 / run	$0.03795 / run	$0 API* / run
Guardrail generated	**Guardrail generated**	**No guardrail generated**

33 author-labelled choices. Review/deployment excluded. *Local compute unpriced; workflow incomplete.
Small pilot; confidence definitions differ. No production superiority claim. Full metrics in backup.

04

Define recovery criteria.

Specify exit tests, approval authority and a reversible transition.



Identify a requirement.
Assign its owner.
Specify acceptance evidence.

Choose one unresolved control boundary in your environment.
Document the evidence needed before approving enforcement.

Patterns to adapt across Kubernetes

Policy, test and incident-exercise source package

Policy templates and fixture tests
Evidence capture and Quarantine levels
Incident cases and lab recordings

Apache-2.0 source. Public repository release pending.
Validate identity, CNI behavior and policy support on your platform.

Fallback: generated policy excerpt

kind: ValidatingPolicy
name: deny-privilege-escalation
validationActions: [Deny]

The policy rejects privileged containers.
Human review approves the generated file before deployment.

Provenance: model = none (spec supplied)

Fallback: fixture admission outcomes

Workload	Policy result
payments-api	Passes privilege and egress checks
backup-job	Would block under the egress-profile policy
report-runner	Registry violation at Audit, admitted
backup-job with exception	Egress check skips the matched exception

Case D: copied credential response

Change	Copied Kubernetes credential	Preserved access
Broad binding	Outside Secret GET/LIST: 200	Named Secret and ConfigMap: 200
Pod quarantined	Outside Secret GET/LIST: 200	Stand-in HTTP health available
RBAC narrowed	Outside Secret GET/LIST: 403	Required named GETs: 200
Old bound Pod deleted	Old credential: 401	Replacement credential: 200

Stand-in health only. External application-key rotation remains required.

Training and classical baselines

Candidate	Correct / 27	Selected / 27	Wrong selected
TF-IDF + logistic regression	20/27	8/27	0
Base Laya	12/27	2/27	0
Trained Laya	12/27	16/27	5

480 synthetic training states. Same task and runtime option order.

Small author-labelled pilot. Confidence needs independent validation.

Setup and total operating cost

Cost component	Hosted API	Local model
Setup	Keys, billing, network, workspace	Runtime, weights, device and memory
Per-run inference	Metered token charges	Owned compute and operations
Shared engineering	Compiler, tests, credentials and observability	Compiler, tests, credentials and observability
Human review	Uncertainty and unmet intent	Uncertainty and unmet intent

One extra minute of review can exceed the API cost savings.

Full comparison: quality and confidence

Metric	Jev 1.13	Opus 5.5	Laya
Correct choices	30/33	32/33	16/33
Decision latency	0.17 s HTTP	2.37 s HTTP	0.09 s in-process
Selected at score 0.6	26/33	33/33	1/33
Wrong among selected	0/26	1/33	0/1
Scenario A time/run	1.82 s	24.18 s	0.41 s
API cost/run	$0.00019	$0.03795	$0 API; compute unpriced

33 scored labels. Small pilot; confidence definitions differ. Laya generated no guardrail.